

Dms - IV Sem Model & Paper Solution

Module 1

Q1) a) Define Tautology. ST $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is a tautology by constructing the truth table.

Defⁿ: Tautology: A compound proposition which is true for all the possible truth value of its components are called Tautology.

→ The following truth table proves the required result

p	q	r	p → r	p ∨ q	q → r	a	b	c
0	0	0	1	0	1	1	0	1
0	0	1	1	0	1	1	0	1
0	1	0	1	1	0	0	1	1
0	1	1	1	1	1	1	1	1
1	0	0	0	1	1	0	1	1
1	0	1	1	1	1	1	1	1
1	1	0	0	1	0	0	1	1
1	1	1	1	1	1	1	1	1

$$a \wedge c = (p \rightarrow r) \wedge (q \rightarrow r)$$

$$b \wedge d = (p \vee q) \wedge \{(p \rightarrow r) \wedge (q \rightarrow r)\}$$

$$c = (p \vee q) \wedge [(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$$

(b) P.T following using the laws of logic

$$[\neg p \wedge (\neg q \wedge r)] \vee [(q \wedge r) \vee (p \wedge r)] \Leftrightarrow r$$

$$\Rightarrow [\neg p \wedge (\neg q \wedge r)] \vee (q \wedge r) \vee (p \wedge r)$$

$$\Leftrightarrow [\neg p \wedge (\neg q \wedge r)] \vee (\sigma \wedge q) \vee (\sigma \wedge p)$$

$$\Leftrightarrow q \wedge p \wedge (\neg q \wedge r) \vee \sigma \wedge (p \vee q)$$

$$\Leftrightarrow [(\neg p \wedge \neg q) \wedge r] \vee (\sigma \wedge (p \vee q))$$

$$\Leftrightarrow [\sigma \wedge (\neg(p \vee q))] \vee [\sigma \wedge (p \vee q)]$$

$$\Leftrightarrow \sigma \wedge [\neg(p \vee q) \vee (p \vee q)]$$

$$\Leftrightarrow \sigma \wedge T \Leftrightarrow \sigma$$

- Q 1) For any two odd integers m, n , ST
- $m+n$ is even
 - mn is odd.

Consider $m = 2k+1$ — odd integer

$$n = 2l+1 \text{ — odd integer}$$

$$(i) \quad m+n = (2k+1) + (2l+1)$$

$$= 2(k+l) + 2 \text{ — even integer}$$

$\therefore m+n$ is even

$$(ii) \quad mn = (2k+1)(2l+1)$$

$$= 2(k+l+kl) + 1 \text{ — even integer}$$

$\therefore mn$ is ~~an~~ odd integer.

- Q 2) Define (i) an open statement
(ii) quantifiers.

(i) Open Statement: Open statements containing a variable x are denoted by $p(x), q(x)$ etc. If U is the universe for the variable x in an open statement $p(x)$ & if $a \in U$, then the proposition got by replacing x by a in $p(x)$ is denoted by $p(a)$.

(ii) Quantifiers: The propositions, which include the words "all", "every", "Some", "there exists" are associated with the idea of a quantity. Such words are called quantifiers.

b) write the following argument in Symbolic form & then establish the validity

If A gets the Supervisor's position & works hard, then he will get a raise. If he gets a raise, then he will buy a car. He has not purchased a car. Therefore he did not get the Supervisor's position or he did not work hard.

→ p : A gets Supervisor's position

q : works hard

r : he gets raise

s : buy a car.

argument is in Symbolic form as

$$(p \wedge q) \rightarrow r$$

$$r \rightarrow s$$

$$\neg s$$

$$\therefore \neg p \vee \neg q \quad \text{or} \quad \neg(p \wedge q)$$

$$\therefore [(p \wedge q) \rightarrow r] \wedge [r \rightarrow s] \wedge (\neg s)$$

$$\Leftrightarrow [(p \wedge q) \rightarrow s] \wedge (\neg s)$$

Rule of Syllogism

Modus Tollens $(p \rightarrow q) \wedge \neg q = \neg p$

$$\Leftrightarrow \neg(p \wedge q)$$

∴ Argument is valid.

C) For the following Statements, the universe comprises all non-zero integers. Determine the truth value of each Statement.

- a) $\exists x \exists y [xy=1]$ b) $\exists x \forall y [xy=1]$
 c) $\forall x \exists y [xy=1]$ d) $\exists x \exists y [(2x+y=5) \wedge (x-3y=-8)]$
 e) $\exists x \exists y [(3x-y=7) \wedge (2x+4y=3)]$

$\Rightarrow$ (i) True. [Take $x=1, y=1$]

(ii) False. (For a specified x , $xy=1$ for every y is not true).

(iii) False. (For $x=2$, there is no integer y such that $xy=1$)

(iv) True. (Take $x=1, y=3$)

(v) False. (Equations $3x-y=7$ & $2x+4y=3$ do not have a common integer solution).

Module 2

Q 3. a) Define the well ordering principle. By mathematical induction, prove that $(n!) \geq \frac{1}{2}n-1$ for all integers $n \geq 1$.

The well ordering principle states as follows:
 Every nonempty subset of $\mathbb{Z}^+$ contains a smallest (least) element.

An alternative way of stating this principle is that the set of all positive integers is well-ordered.

Here we have prove that

$S(n) : n! \geq 2^{n-1}$ is true for all integers $n \geq 1$

Basis Step: For $n=1$, $S(n)$ reads $1! \geq 2^{1-1}$, which is obviously true.

Thus $S(n)$ is verified for $n=1$

Induction Step: we assume that $S(n)$ is true for $n=k$, where $k \geq 1$; that is, we assume that

$$k! \geq 2^{k-1} \text{ or } 2^{k-1} \leq k!$$

is true, This yields

$$2^k = 2 \cdot 2^{k-1} \leq 2 \cdot k!$$

$$< (k+1) \cdot k! \quad \because 2 < k+1 \text{ for } k \geq 1$$

$$= (k+1)!$$

$$\text{ie } (k+1)! > 2^k.$$

This is precisely the statement $S(n)$ for $n=k+1$.

Thus, on the assumption that $S(n)$ is true for $n=k \geq 1$, we have proved that $S(n)$ is true for $n=k+1$.

Hence, by mathematical induction, $S(n)$ is true for all integers $n \geq 1$.

b) p.t. every positive integer $n \geq 24$ can be written as a sum of 5's & 7's.

$\Rightarrow$ Hence, we have to prove that the statement $S(n)$: n can be written as a sum of 5's and 7's is true for all integers $n \geq 24$.

Basis Step: we note that

$$24 = (7+7) + (5+5)$$

This shows that $S(24)$ is true.

Induction Step: we assume that $S(n)$ is true for $n = k$, where $k \geq 24$.

$$\text{Then } k = (7+7+\dots) + (5+5+\dots)$$

Suppose this representation of k has r no. of 7's and s no. of 5's. Since $k \geq 24$, we should have $r \geq 2$ & $s \geq 2$.

Using this representation of k , we find that

$$k+1 = \{ (7+7+\dots) + (5+5+\dots) \} + 1$$

$$= \underbrace{(7+7+\dots)}_{(r-2)} + \underbrace{(7+7)}_s + \underbrace{(5+5+\dots)}_s + 1$$

$$= \underbrace{(7+7+\dots)}_{(r+2)} + \underbrace{(5+5+\dots)}_{s+3}$$

This shows that $(k+1)$ is a sum of 7's & 5's.

Thus $S(k+1)$ is true.

c) How many positive integers n , can we form using the digits 3, 4, 4, 5, 5, 6, 7, if we want n to exceed 5,000,000?

→ Here n must be of the form

$$n = x_1 x_2 x_3 x_4 x_5 x_6 x_7$$

where $x_1, x_2, \dots, x_7$ are the given digits with

$x_1 = 5, 6$ or 7 . Suppose we take $x_1 = 5$. Then $x_2 x_3 x_4 x_5 x_6 x_7$ is an arrangement of the remaining 6 digits which contains two 4's and one each of 3, 5, 6, 7. The no^r of such arrangements is

$$\frac{6!}{2!1!1!1!1!1!} = 360$$

Next, Suppose we take $x_1 = 6$. Then $x_2 x_3 x_4 x_5 x_6 x_7$ is an arrangement of 6 digits which contains two each of 4 & 5 and one each of 3 & 7. The no^r of such arrangements is

$$\frac{6!}{2!2!1!1!} = 180$$

Similarly, if we take $x_1 = 7$, the no^r of arrangements is

$$\frac{6!}{2!2!1!1!} = 180$$

Accordingly, by the Sum rule, the no^r of n 's of the desired type is

$$360 + 180 + 180 = 720$$

Q4) a) By Mathematical induction, prove that
 $1 \cdot 3 + 2 \cdot 4 + \dots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$

$\Rightarrow$ Hence, we have to prove the Statement

$$S(n) = 1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + \dots + n(n+2) = \frac{1}{6} n(n+1)(2n+7), \quad n \geq 1$$

Basis Step: we note that $S(1)$ is the Statement

$$1 \cdot 3 = \frac{1}{6} \cdot 1(1+1) \cdot (2+7)$$

which is clearly true. Thus, the Statement $S(n)$ is verified for $n=1$.

Induction Step: we assume that the Statement $S(n)$

is true for $n=k$, where k is an integer ≥ 1 ; that is,

we assume that the following Statement is true:

$$S(k) = 1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + \dots + k(k+2) = \frac{1}{6} k(k+1)(2k+7)$$

Using this we find that (by adding $(k+1)(k+3)$ both side)

$$S(k) = 1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + \dots + k(k+2) + (k+1)(k+3) = \frac{1}{6} k(k+1)(2k+7) + (k+1)(k+3)$$

$$= (k+1) \left\{ \frac{1}{6} k(2k+7) + (k+3) \right\}$$

$$= (k+1) [2k^2 + 7k + 6k + 18]$$

$$= (k+1)(2k^2 + 13k + 18)$$

$$= \frac{1}{6} (k+1)(k+2)(2k+9)$$

This is precisely the Statement $S(k+1)$. Thus, on the basis of the assumption that $S(n)$ is true for $n=k \geq 1$.

Thus the truthness of $S(n)$ for $n=k+1$ is established.

b) Find the no of permutations of the letters of the word MASSASAUGA. In how many of these all four A's are together? How many of them begin with S?

→ The given word has 10 letters of which 4 are A, 3 are S and 1 each are M, U and G.

∴ The required no of permutations is

$$\frac{10!}{4!3!1!1!1!} = 25200$$

If, in a permutation, all A's are to be together, we treat all of A's as one single letter. Then the letters to be permuted are (AAAA), S, S, S, M, U, G (which are 7 in no), & the no of permutations is

$$\frac{7!}{1!3!1!1!1!} = 840.$$

For permutations beginning with S, there occur nine open positions to fill, where two are S, four are A, & one each are M, U, G. The no of each such permutations is

$$\frac{9!}{2!4!1!1!1!} = 7560.$$

(c) (i) Obtain the coefficient of a^5b^2 in the expansion of $(2a+3b)^7$

(ii) Using the Binomial theorem find the coefficient of x^5y^2 in the expansion of $(x+y)^7$.

(i) we have, by the binomial thm

$$(2a-3b)^7 = \sum_{r=0}^7 \binom{7}{r} (2a)^r (-3b)^{7-r}$$

$$= \sum_{r=0}^7 \binom{7}{r} 2^r (-3)^{7-r} a^r b^{7-r}$$

in the expansion, the coefficient of $a^5 b^2$ (which corresponds to $r=5$) is

$$\binom{7}{5} 2^5 (-3)^2 = 6048$$

(ii) we have, by the binomial thm

$$(x+y)^7 = \sum_{r=0}^7 \binom{7}{r} x^r y^{7-r}$$

$$= \sum_{r=0}^7 \binom{7}{r} x^r y^{7-r}$$

in the expansion, the coefficient of $x^5 y^2$ (which corresponds to $r=5$) is

$$\binom{7}{5} 1^5 1^2 = 21$$

Module - 3

Q 5) a) State pigeon-hole principle. p.T if any no^s from 1 to 8 are chosen then two of them have their sum is 9.

⇒ If m pigeons occupy n pigeon holes and if $m > n$; then two or more pigeons occupy the same pigeonhole.

Let us first write all the nos from 1 to 8
 $\{1, 2, 3, 4, 5, 6, 7, 8\}$

Now let's take any 5 nos from 1 to 8 such as $\{1, 3, 4, 7, 8\}$

As it is given that any two of the nos out of the 5 nos we have chosen should be equal to sum 9.

Let's add every two nos

(case i) $1 + 3 = 4$

ii) $3 + 4 = 7$

iii) $4 + 7 = 11$

iv) $7 + 8 = 15$

v) $8 + 1 = 9$

In case 5 we get a pair of nos

So according to the principle of pigeonhole,

we can take any 5 nos & there will

always exist one pair whose sum is equal to 9.

b) let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by, $f(x) = \begin{cases} 3x-5, & \text{if } x > 0 \\ 1-3x, & \text{if } x \leq 0 \end{cases}$
 find, $f^{-1}([-6, 5])$ & $f^{-1}([-5, 5])$

$$\Rightarrow f^{-1}([-6, 5]) = \{x \in \mathbb{R} \mid f(x) \in [-6, 5]\}$$

$$= \{x \in \mathbb{R} \mid -6 \leq f(x) \leq 5\}$$

when $x > 0$, we have, $f(x) = 3x-5$, $\therefore -6 \leq f(x) \leq 5$

whenever, $-6 \leq (3x-5) \leq 5$, or $-1 \leq 3x \leq 10$
or $-\frac{1}{3} \leq x \leq \frac{10}{3}$

when $x \leq 0$, we have, $f(x) = -3x+1$

$\{x \in \mathbb{R} \mid -6 \leq f(x) \leq 5\}$, whenever $-6 \leq (-3x+1) \leq 5$
or $-7 \leq -3x \leq 4$
or $-\frac{7}{3} \leq x \leq -\frac{4}{3}$

Thus, $f^{-1}([-6, 5]) = \{x \in \mathbb{R} \mid -\frac{7}{3} \leq x \leq -\frac{4}{3} \text{ or } -\frac{1}{3} \leq x \leq \frac{10}{3}\}$
 $= \{x \in \mathbb{R} \mid -\frac{7}{3} \leq x \leq \frac{10}{3}\}$

$\therefore f^{-1}([-6, 5]) = [-\frac{7}{3}, \frac{10}{3}]$

H4 $f^{-1}([-5, 5]) = \{x \in \mathbb{R} \mid f(x) \in [-5, 5]\}$
 $= \{x \in \mathbb{R} \mid -5 \leq f(x) \leq 5\}$

when $x > 0$, we have, $f(x) = 3x-5$, $\therefore -5 \leq f(x) \leq 5$

whenever $-5 \leq 3x-5 \leq 5$ or $0 \leq 3x \leq 10$ or
 $0 \leq x \leq \frac{10}{3}$

when $x \leq 0$, we have $f(x) = -3x + 1$.

$$\therefore -5 \leq f(x) \leq 5, \text{ whenever } -5 \leq (-3x+1) \leq 5$$
$$-6 \leq -3x \leq 4$$

$$\underline{\text{or}} \quad 2 \geq x \geq -4/3 \quad \underline{\text{or}} \quad -4/3 \leq x \leq 2$$

$$\text{Thus, } f^{-1}([-5, 5]) = \{x \in \mathbb{R} \mid -\frac{4}{3} \leq x \leq 2 \text{ or } 0 \leq x \leq \frac{10}{3}\}$$
$$= \{x \in \mathbb{R} \mid -\frac{4}{3} \leq x \leq 10/3\}$$
$$= \{-4/3, 10/3\}$$

~~we~~ we find that $\boxed{f^{-1}[-5, 5] = \{-4/3, 10/3\}}$

(C) Let $A = B = C = \mathbb{R}$ & $f: A \rightarrow B$ & $g: B \rightarrow C$ be defined by

$$f(a) = 2a + 1, \quad g(b) = \frac{1}{3}b, \quad \forall a \in A, \forall b \in B$$

Compute $g \circ f$ & s.t. $g \circ f$ is invertible.

what is $(g \circ f)^{-1}$?

$$\rightarrow \text{we have } (g \circ f)(a) = g(f(a)) = g(2a+1) = \frac{1}{3}(2a+1)$$

$$\text{Thus, } g \circ f: A \rightarrow C \text{ is defined by } (g \circ f)(a) = \frac{1}{3}(2a+1)$$

we check that f is invertible with $f^{-1}(b) = \frac{1}{2}(b-1)$ & g is invertible with $g^{-1}(c) = 3c$.

$\therefore g \circ f$ is invertible, and its inverse is given by

$$(g \circ f)^{-1}(c) = (f^{-1} \circ g^{-1})(c)$$

$$= f^{-1}\{g^{-1}(c)\} = f^{-1}(3c)$$

$$= \frac{1}{2}(3c-1).$$

Q) 6) a) Let f and g be funcs from $\mathbb{R}$ to $\mathbb{R}$ defined by
 $f(x) = ax + b$ & $g(x) = 1 - x + x^2$, if $(g \circ f)(x) = 9x^2 - 9x + 3$
 determine a & b .

$\Rightarrow$ we have, $9x^2 - 9x + 3 = (g \circ f)(x)$

$$9x^2 - 9x + 3 = g\{f(x)\} = g(ax + b)$$

$$9x^2 - 9x + 3 = [1 - (ax + b) + (ax + b)^2]$$

$$= a^2x^2 - 2abx + b^2 + 1 - ax - b$$

$$= a^2x^2 + (2ab - a)x + (1 - b + b^2)$$

Comparing the corresponding Coefficient we get

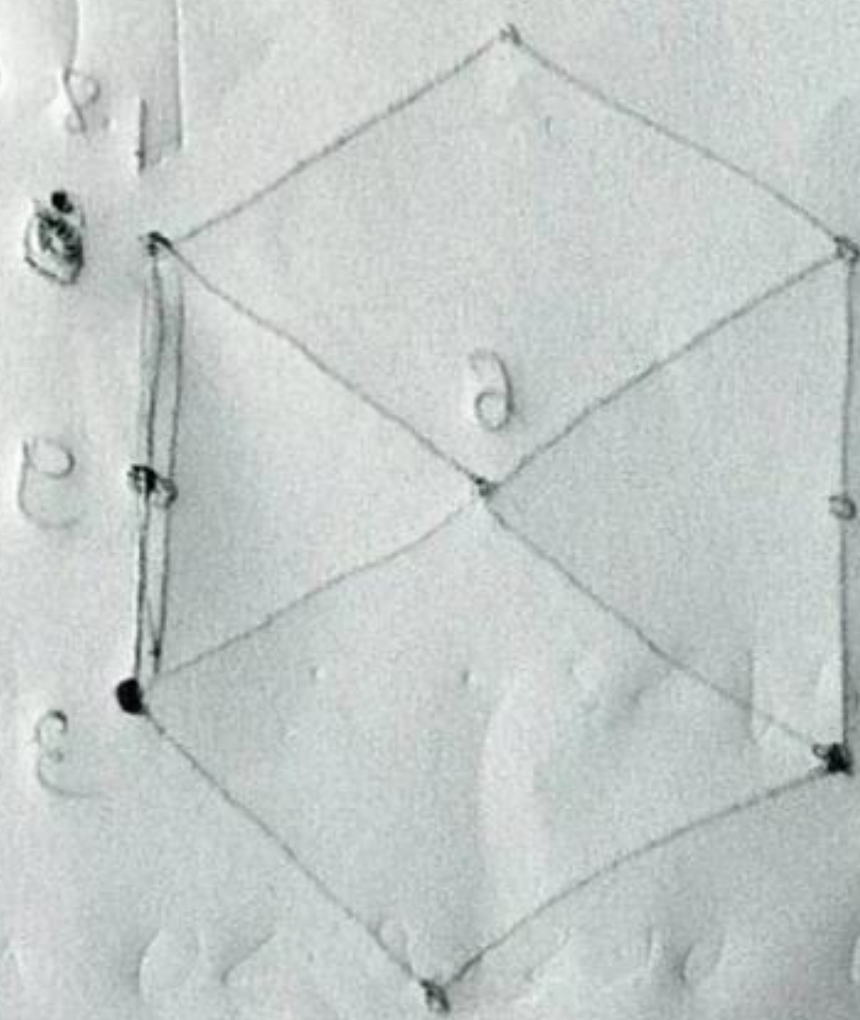
$$9 = a^2, \quad 2ab - a = -9, \quad 1 - b + b^2 = 3$$

$$a = \pm 3, \quad \text{for } a = 3, \quad 6b - 3 = -9$$

$$\boxed{b = -1}$$

$$\text{for } a = -3, \quad b = 2$$

Thus, $\boxed{a = 3 \quad b = -1}$ & $\boxed{a = -3 \quad b = 2}$ are the
 values of a & b .



b) Draw the Hasse (poset) diagram which represents positive divisors of 36.

→ The set of +ve divisors of 36 is

$D_{36} = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$ The Relation R of divisibility (i.e. aRb iff a divides b) is a partial order ~~set~~ on set. The Hasse diagram of this partial order is required here

1 is related all elements of D_{36}

2 " " to 2, 4, 6, 12, 18, 36

3 " " 3, 6, 9, 12, 18, 36

4 " " 4, 12, 36

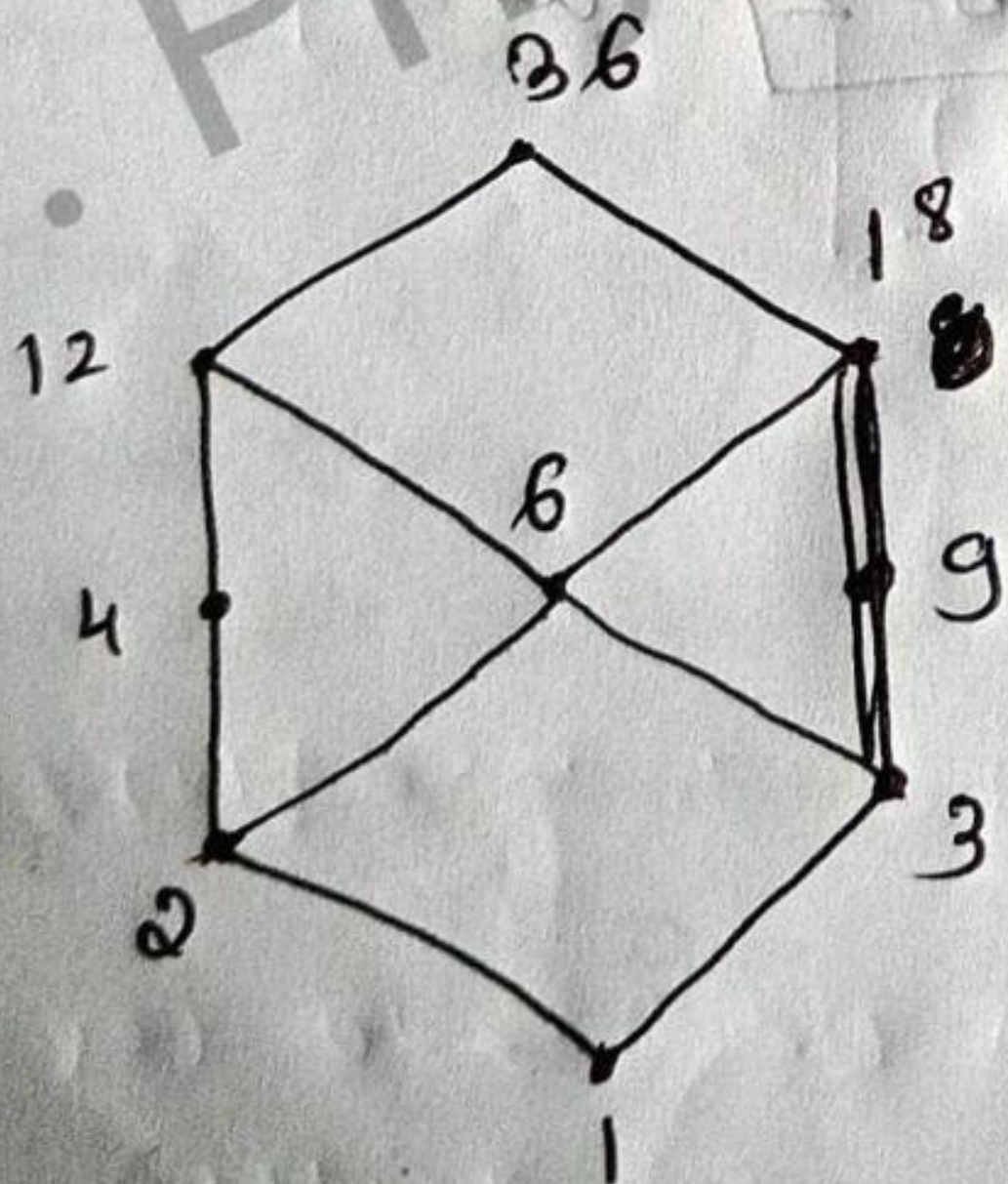
6 " " 6, 12, 18, 36

9 " " 9, 18, 36

12 " " 12, 36

18 " " 18, 36

36 " " 36



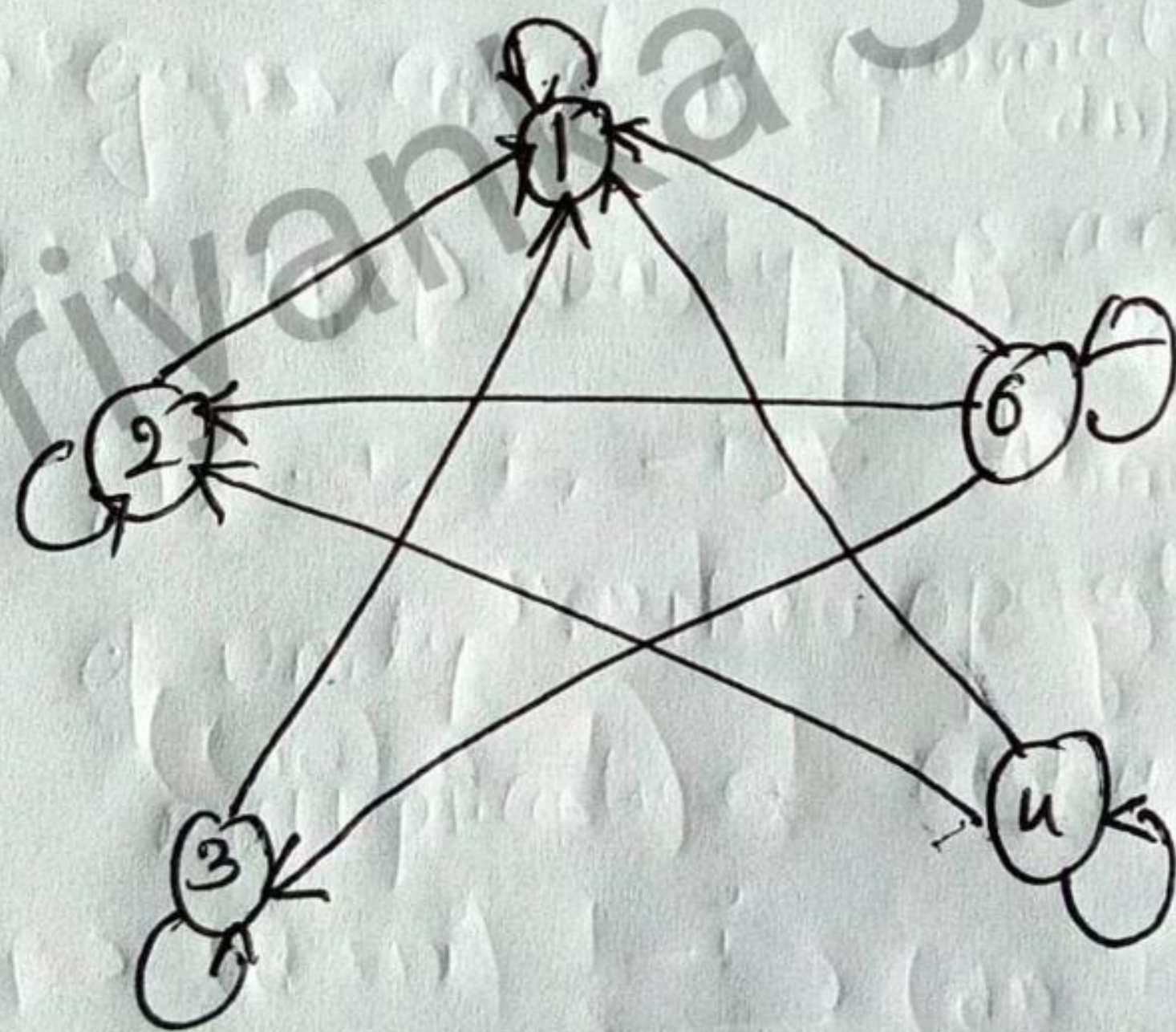
C) Let $A = \{1, 2, 3, 4, 6\}$ & R be a relation on A defined by aRb iff " a is multiple of b ". Write down the relation R , relation matrix $M(R)$ & draw its digraph. List out its in degree & out degree.

$$\rightarrow R = \{(1, 1), (2, 1), (2, 2), (3, 1), (3, 3), (4, 1), (4, 2), (4, 4), (6, 1), (6, 2), (6, 3), (6, 6)\}$$

By examining the elements of R , we find the matrix R is $M_R =$

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

The digraph of R is as shown below



Vertex	1	2	3	4	6
In-degree	4	3	2	0	1
Out-degree	1	2	2	3	4

Module - 4

Q.7 a) Determine the no^r of positive integers n such that $1 \leq n \leq 100$ and n is not divisible by 2, 3 or 5.

$\Rightarrow$ Let $S = \{1, 2, 3, \dots, 100\}$. Then $|S| = 100$. Let A_1, A_2, A_3 be the subsets of S whose elements are divisible by 2, 3 and 5 respectively. Then, we have to find $|\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}|$.

We note that, $|A_1| = \text{No. of elements in } S \text{ that are divisible by 2}$

$$= \lfloor 100/2 \rfloor = \lfloor 50 \rfloor = 50,$$

$|A_2| = \text{No. of elements in } S \text{ that are divisible by 3}$

$$= \lfloor 100/3 \rfloor = \lfloor 33.333 \rfloor = 33,$$

$|A_3| = \text{No. of elements in } S \text{ that are divisible by 5}$

$$= \lfloor 100/5 \rfloor = \lfloor 20 \rfloor = 20,$$

$|A_1 \cap A_2| = \text{No. of elements in } S \text{ that are divisible by 2 \& 3}$

$$= \lfloor 100/6 \rfloor = \lfloor 16.666 \rfloor = 16,$$

$|A_2 \cap A_3| = \text{No. of elements in } S \text{ that are divisible by 3 and 5}$

$$= \lfloor 100/15 \rfloor = \lfloor 6.666 \rfloor = 6,$$

$|A_1 \cap A_3| = \text{No. of elements in } S \text{ that are divisible by 2 and 5}$

$$= \lfloor 100/10 \rfloor = \lfloor 10 \rfloor = 10,$$

$|A_1 \cap A_2 \cap A_3| = \text{No. of elements in } S \text{ that are divisible by 2, 3, and 5}$

$$= \lfloor 100/30 \rfloor = \lfloor 3.333 \rfloor = 3.$$

Now, the principle of inclusion-exclusion gives

(see

$$|\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}| = |S| - \{|A_1| + |A_2| + |A_3|\} + \{|A_1 \cap A_2| + |A_1 \cap A_3| + |A_2 \cap A_3|\} - |A_1 \cap A_2 \cap A_3|$$

$$= 100 - (50 + 33 + 20) + (16 + 10 + 6) - 3 = 26$$

Thus, the required no. is 26.

6) In how many ways can the 26 letters of the english alphabet be permuted so that none of the patterns CAR, DOG, PUN or BYTE occurs?

⇒ let S denote the set of all permutations of the 26 letters.

Then $|S| = 26!$.

let A_1 be the set of all permutations in which CAR appears. This word, CAR, consists of three letters which form a single block. The set A_1 therefore consists of all permutations which contain this single block and the 23 remaining letters.

$$|A_1| = 24!$$

Similarly, if A_2, A_3, A_4 are the sets of all permutations which contain DOG, PUN and BYTE respectively, we have

$$|A_2| = 24!, \quad |A_3| = 24!, \quad |A_4| = 23!$$

Like wise we find that

$$|A_1 \cap A_2| = |A_1 \cap A_3| = |A_2 \cap A_3| = (26 - 2 + 2)! = 22!$$

$$|A_1 \cap A_4| = |A_2 \cap A_4| = |A_3 \cap A_4| = (26 - 7 + 2) = 21!$$

$$|A_1 \cap A_2 \cap A_3| = (26 - 9 + 3)! = 20!$$

$$|A_1 \cap A_2 \cap A_4| = |A_1 \cap A_3 \cap A_4| = |A_2 \cap A_3 \cap A_4| = (26 - 10 + 3)! = 19!$$

$$|A_1 \cap A_2 \cap A_3 \cap A_4| = (26 - 13 + 4)! = 17!$$

∴ The required no. of permutations is given by

$$|\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \cap \bar{A}_4| = |S| - \sum |A_i| + \sum |A_i \cap A_j| - \sum |A_i \cap A_j \cap A_k| + |A_1 \cap A_2 \cap A_3 \cap A_4|$$

$$= 26! - (3 \times 24! + 3 \times 23!) + (3 \times 22! + 3 \times 21!) - (20! + 3 \times 19!) + 17!$$

③ Solve the recurrence relation $a_n = n a_{n-1}$, where $n \geq 1$ & $a_0 = 1$

→ From the given relation, we find that

$$a_1 = 1 \times a_0, a_2 = 2 a_1 = (2 \times 1) a_0$$

$$a_3 = 3 a_2 = (3 \times 2 \times 1) a_0$$

$$a_n = n a_{n-1} = (n \times 3 \times 2 \times 1) a_0 \text{ \& so on}$$

Evidently, the general solution is (by induction)

$$a_n = (n!) a_0, \text{ for } n \geq 1$$

Using the given initial condition $a_0 = 1$ in this we get $a_n = n!$ as the required solution.

Q 8) a) In how many ways 5 no. of a's, 4 no. of b's and 3 no. of c's can be arranged so that all the identical letters are not in a single block?

→ The given letters are $5 + 4 + 3 = 12$ in no., of which

5 are a's 4 are b's and 3 are c's.

If S is the set of all permutations (arrangements) of these letters, we have,

$$|S| = \frac{12!}{5! 4! 3!}$$

Let A_1 be the set of arrangements of the letters a, a, a, a, a where the 5 a's are in a single block.

The no. of such arrangement is

$$|A_1| = \frac{8!}{4!3!}$$

[$\because$ in such an arrangement all the a's taken together can be regarded as a single letter & the remaining letters consist of 4 b's & 3 c's]

III, if A_2 is the set of arrangements where the 4 b's are in a single block, & A_3 is the set of arrangements where the 3 c's are in a single block, we have

$$|A_2| = \frac{9!}{5!3!} \quad \& \quad |A_3| = \frac{10!}{5!4!}$$

$$\text{Likewise, } |A_1 \cap A_2| = \frac{5!}{3!}, \quad |A_1 \cap A_3| = \frac{6!}{4!},$$

$$|A_2 \cap A_3| = \frac{7!}{5!},$$

$$|A_1 \cap A_2 \cap A_3| = 3!$$

Accordingly, the required no. of arrangements is

$$|\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}| = |S| - \{|A_1| + |A_2| + |A_3|\} + \{|A_1 \cap A_2| + |A_1 \cap A_3| + |A_2 \cap A_3|\} - |A_1 \cap A_2 \cap A_3|$$

$$= \frac{10!}{5!4!3!} - \left\{ \frac{8!}{4!3!} + \frac{9!}{5!3!} + \frac{10!}{5!4!} \right\} + \left\{ \frac{5!}{3!} + \frac{6!}{4!} + \frac{7!}{5!} \right\} - 3!$$

$$= 24720 - (280 + 504 + 1260) + (20 + 30 + 42) - 6$$

$$= 25762 //$$

b) Five teachers T_1, T_2, T_3, T_4, T_5 are to be made class teachers for five classes, C_1, C_2, C_3, C_4, C_5 , one teacher for each class. T_1 and T_2 do not wish to become the class teachers for C_1 or C_2 . T_3 and T_4 for C_4 or C_5 , and T_5 for C_3 or C_4 or C_5 . In how many ways can the teachers be assigned the work (without displeasing any teacher)?

→ The situation can be represented by the board shown below in which the rows represent respectively the teachers T_1, T_2, T_3, T_4, T_5 and the columns represent the classes C_1, C_2, C_3, C_4, C_5 , and the shaded squares together represent the forbidden places involved.

	C_1	C_2	C_3	C_4	C_5
T_1	shaded	shaded			
T_2	shaded	shaded			
T_3				shaded	shaded
T_4				shaded	shaded
T_5			shaded	shaded	shaded

For the board C made up of the shaded squares in the above figure, the look polynomial is given by

$$\sigma(C, x) = 1 + 11x + 40x^2 + 56x^3 + 28x^4 + 4x^5$$

Thus,

$$\text{here } \sigma_1 = 11, \sigma_2 = 40$$

$$\sigma_3 = 56, \sigma_4 = 28$$

$$\sigma_5 = 4$$

Consequently,

$$S_0 = 5! = 120, S_1 = (5-1)! \times \sigma_1 = 264$$

$$S_2 = (5-2)! \times \sigma_2 = 240$$

$$S_3 = (5-3)! \times \sigma_3 = 112$$

$$S_4 = (5-4)! \times \sigma_4 = 28$$

$$S_5 = (5-5)! \times \sigma_5 = 4$$

Accordingly, the no^r of ways in which the work can be assigned is

$$\begin{aligned} S_0 - S_1 + S_2 - S_3 + S_4 - S_5 &= 120 - 264 + 240 - 112 + 28 - 4 \\ &= 8 \end{aligned}$$

89) Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$
when $n \geq 0$ & $F_0 = 0$ & $F_1 = 1$.

when rewritten, the given relation reads

$$F_n - F_{n-1} - F_{n-2} = 0 \quad \text{for } n \geq 2.$$

The characteristic equation for this is $k^2 - k - 1 = 0$ whose roots are

$$k = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}.$$

According, the general solution for F_n is

$$F_n = A \left(\frac{1+\sqrt{5}}{2} \right)^n + B \left(\frac{1-\sqrt{5}}{2} \right)^n \quad \text{--- (i)}$$

where A and B are arbitrary constants. Using the given initial conditions $F_0 = 0$ & $F_1 = 1$ in this we get

$$0 = A \left(\frac{1+\sqrt{5}}{2} \right)^0 + B \left(\frac{1-\sqrt{5}}{2} \right)^0 = A + B$$

$$1 = A \left(\frac{1+\sqrt{5}}{2} \right) + B \left(\frac{1-\sqrt{5}}{2} \right)$$

Solving these, we get $A = -B = \frac{1}{\sqrt{5}}$. Putting these values of A and B into (i), we get

$$F_n = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right\}$$

as the solution for F_n with $F_0 = 0$ & $F_1 = 1$

Module - 5

Q9. a) Define group. S.T fourth roots of unity is an abelian group.

If G is a non-empty set & $*$ is a binary operation on G , then $(G, *)$ is called a group. If the following conditions are satisfied:

- 1) $\forall a, b \in G, a * b \in G$ [closure of G under $*$]
 - 2) $\forall a, b, c \in G, a * (b * c) = (a * b) * c$ [associative]
 - 3) $\exists e \in G$ with $a * e = e * a = a, \forall a \in G$ [existence of identity]
 - 4) For each $a \in G$ there is an element $b \in G$ such that $a * b = b * a = e$ [existence of inverse]
- $\pm b$, in addition $a * b = b * a$ for all $a, b \in G$. Then G is called a commutative or abelian group.

→ fourth roots of unity is an abelian gp

Let $W_4 = \{1, -1, i, -i\}$, the set of all fourth roots of unity. The operation Table for the usual multiplication on W_4 is as shown below

X	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

From the table, it is evident that w_4 is closed under $\times$. Since $\times$ is associative in the set of complex nos it is associative in w_4 . Further, 1 is in w_4 & is the identity element for $\times$. Also, every element of A has an inverse under $\times$ (examine the table). Lastly $\times$ is commutative.

Hence w_4 is an abelian group under $\times$.

This group is a finite group of order 4.

(b) If G be a set of all non-zero real nos & let $a * b = ab/2$ then $(G, *)$ is an abelian gp.

$\Rightarrow$ For any two non-zero real nos a & b we note that $\frac{1}{2}(ab)$ is a non-zero real no.

This means that, for any $a, b \in G$, $a * b \in G$.

$\therefore G$ is closed under $*$.

For any $a, b, c \in G$, we have

$$\begin{aligned} a * (b * c) &= a * \left(\frac{1}{2}bc\right) = \frac{1}{2} \left(\frac{1}{2}a(bc)\right) = \frac{abc}{4} \\ &= \frac{1}{2} \left(\frac{1}{2}ab\right) c = \frac{1}{2} (a * b) c = (a * b) * c \end{aligned}$$

Thus, $*$ is associative.

For any $a \in G$ we have

$$a * 2 = 2 * a = \frac{1}{2} (2a) = a$$

$\therefore 2$ is the identity under $*$, & $2 \in G$

For any $a \in G$, if we set $a' = 4/a$, then $a' \in G$ &

$$a * a' = \frac{1}{2} (aa') = \frac{1}{2} \cdot 4 = 2 = \text{Identity}$$

$$a' * a = \frac{1}{2} (a'a) = \frac{1}{2} \cdot 4 = 2 = \text{Identity}$$

Thus, every a in G has $a' = 4/a$ as the inverse in G under $*$.

The above facts prove that $(G, *)$ is a group

Further, for any $a, b \in G$

$$a * b = \frac{1}{2} (ab) = \frac{1}{2} (ba) = b * a$$

$\therefore$ The group $(G, *)$ is an abelian group.

(C) Define Klein 4-group. And if $A = \{e, a, b, c\}$
 then S.T. this is a Klein-4 group.

→ Any group of order 4 is called Klein 4-group
 If every element of a group are self inverse.

Let $A = \{e, a, b, c\}$ is. Then each non-identity
 in A is of order 2
 i.e. $a^2 = e$, $b^2 = e$, $c^2 = e$.

By multiplying a^{-1} , b^{-1} , c^{-1} from left sides of
 the above eqs, we obtain that

$$a^{-1}a^2 = a^{-1}e, \quad b^{-1}b^2 = b^{-1}e, \quad c^{-1}c^2 = c^{-1}e$$

Using associative of the group operation

$$(a^{-1}a)a = a^{-1}e, \quad (b^{-1}b)b = b^{-1}e, \quad (c^{-1}c)c = c^{-1}e$$

$$\therefore ea = a^{-1}e, \quad eb = b^{-1}e, \quad ec = c^{-1}e$$

Now using the defⁿ of identity element

$$\text{we have } a = a^{-1}, \quad b = b^{-1}, \quad c = c^{-1}$$

Now we can conclude that every element
 in the group is equal to its own inverse.

→ The Klein 4 group's table is

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

1. Identity element: e

2. Inverse, $e \cdot e = e$

$$a \cdot a = e$$

$$b \cdot b = e$$

$$c \cdot c = e$$

each element its own inverse

3. Group X^n .

$$a \cdot b = c$$

$$a \cdot c = b$$

$$b \cdot c = a$$

$$b \cdot a = c$$

$$c \cdot a = b$$

$$c \cdot b = a$$

Commutative property

* Commutative $a \cdot b = b \cdot a$ etc

• Each element squared is the identity

$$a^2 = b^2 = c^2 = e //$$

Q. 10) a) Define cyclic gp & S.T (eg, 8) whose X^n table is given below is cyclic.

$\cdot$	a	b	c	d	e	f
a	a	b	c	d	e	f
b	b	c	d	e	f	a
c	c	d	e	f	a	b
d	d	e	f	a	b	c
e	e	f	a	b	c	d
f	f	a	b	c	d	e

→ ~~we note that a is the~~

Defⁿ: A group $(G, *)$ is said to be a

cyclic group if $\exists$ an element $a \in G$

such that $G = \{a^n \mid n \in \mathbb{Z}\}$. a is said to be generator of the cyclic group. The cyclic group generated by a is denoted by $\langle a \rangle$. i.e. $\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$

$\Rightarrow$ We note that a is the identity element of G .

we find that,

$$b^2 = b * b = c$$

$$b^3 = b^2 * b = c * d = d$$

$$b^6 = b^5 * b = f * b = a$$

$$b^4 = b^3 * b = d * b = e$$

$$b^5 = b^4 * b = e * b = b$$

Thus every element of G is an integral power of b .

$\therefore (G, *)$ is a Cyclic group with b as a generator.

(b) State & prove Lagrange's thm.

Thm: If G is a finite group & H is a subgroup of G , then the order of H divides the order of G .

Prf: Since G is a finite group, H is finite.

$\therefore$ the no. of cosets of H in G is finite. Let $Ha_1, Ha_2, \dots, Ha_r$ be the distinct

right cosets of H in G . Then by the right coset decomposition of G we have

$$G = Ha_1 \cup Ha_2 \cup Ha_3 \cup \dots \cup Ha_r$$

$$\text{So that } o(G) = o(Ha_1) + o(Ha_2) + \dots + o(Ha_r)$$

$$\text{But } o(Ha_1) = o(Ha_2) = \dots = o(Ha_r) = o(H)$$

$$1. \quad o(e_g) = o(H) + o(H) + o(H) + \dots + o(H)$$

$$o(e_g) = \cancel{2 \times o(H)} \quad 2 \times o(H)$$

→ This shows that $o(H)$ divides $o(e_g)$

$$\Rightarrow o(H) \mid o(e_g)$$

→ This proof of the theorem is complete

(C) Let G be a group with subgroup H & K
 ~~$H \cap K = \{e\}$~~ Let $|G| = 660$ & $|K| = 66$ & $K \cap H \subseteq G$
 and find the possible value for $|H|$

→ By Lagrange's thm, $|H|$ must divide $|G| = 660$

& $|K| = 66$ must divide $|H|$ also

we have $|G| \neq |H| \neq |K|$

$\therefore 660 = |H|q_1$ for some integer $q_1 > 1$

& $|H| = 66q_2$ for some integer $q_2 > 1$

Hence $660 = 66q_1q_2$ or $q_1q_2 = 10$

Thus, either $q_1 = 2$ & $q_2 = 5$

or $q_1 = 5$ & $q_2 = 2$

→ Thus, $|H| = 66 \times 5 = 330$

or $|H| = 66 \times 2 = 132$